

Programming Using C++ Level 1

Nested For Loops



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Definition

- ◆ A nested for loop is a for loop placed inside another for loop.
- ◆ The inner loop runs completely for every single iteration of the outer loop.
- ◆ Nested loops are useful when working with combinations, rows and columns, or multi-dimensional data.

Explanation

- ◆ The outer loop controls the first letter, from A to Z.
- ◆ The inner loop controls the second letter, also from A to Z.
- ◆ For every value of i , the inner loop runs from A to Z.
- ◆ This produces all possible two-letter combinations.

Steps in main()

1. Start the outer loop

◆ Set i to A and continue until i reaches Z .

2. Start the inner loop

◆ For each i , set j to A and continue until j reaches Z .

3. Print the combination

◆ Print i followed by j .

4. Finish the inner loop

◆ After j reaches Z , the inner loop ends.

5. Move to the next outer value

◆ Increase i and start the inner loop again from A .

6. Continue until ZZ

◆ The loops stop after all combinations have been printed.

💡 Example

- ◆ When: $i = A$
- ◆ The inner loop produces:
- ◆ Then i becomes B :
- ◆ The process continues until:

➡ Result

- ◆ The program starts with:

AA

AB

AC

...

AZ

- ◆ Then:

BA

BB

BC

...

BZ

- ◆ And continues until the final combination:

ZZ

- ◆ The program also prints a separator after each complete inner loop.

Note

- ◆ char variables can be used as loop counters because characters have numeric values internally.
- ◆ The loop from 'A' to 'Z' works because the character values are ordered.
- ◆ The outer loop runs 26 times.
- ◆ The inner loop runs 26 times for each outer iteration.
- ◆ Therefore, the program generates $26 \times 26 = 676$ combinations.

⊘ Warning

- ◆ Nested loops can perform many iterations when the number of elements increases.
- ◆ Always make sure both loops have conditions that eventually become false.



Comparison

▨ Nested For Loop

- ✓ Uses one loop inside another loop.
- ✓ The inner loop completes all its iterations for each outer-loop iteration.
- ✓ Useful for generating combinations and processing multiple dimensions.

▤ Single For Loop

- ✓ Controls only one sequence of repetition.
- ✓ Does not naturally handle two independent values.
- ✓ Simpler when only one counter is needed.

✦✦ Summary

- ✓ A nested for loop contains one loop inside another loop.
- ✓ The inner loop runs completely for every iteration of the outer loop.
- ✓ Nested loops are useful for generating combinations and processing multiple levels of data.

Thank you 😊